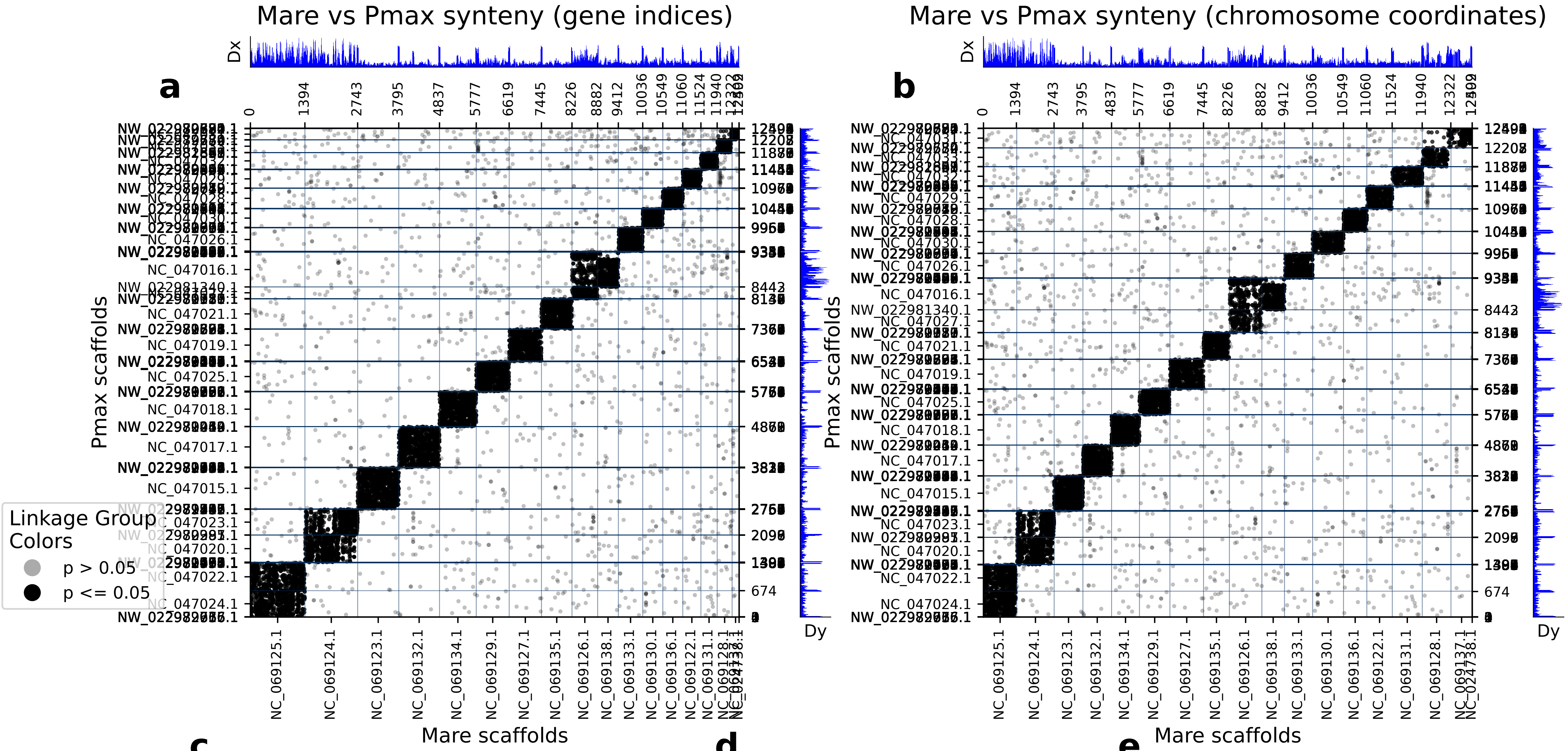
diamond: 12503 orthologs on 18 Mare scaffolds and 212 Pmax scaffolds.



c

Mare vs Pmax groupings

d

Mare vs Pmax FET, whole chroms.

	Mare scaf	Pmax scaf	p (FET)	Count
0	NC_069123.1	NC_047015.1	0.00e+00	958
1	NC_069132.1	NC_047017.1	0.00e+00	930
2	NC_069134.1	NC_047018.1	0.00e+00	824
3	NC_069127.1	NC_047019.1	0.00e+00	735
4	NC_069135.1	NC_047021.1	0.00e+00	685
5	NC_069129.1	NC_047025.1	0.00e+00	681
6	NC_069125.1	NC_047022.1	0.00e+00	623
7	NC_069124.1	NC_047020.1	0.00e+00	614
8	NC_069125.1	NC_047024.1	0.00e+00	592
9	NC_069124.1	NC_047023.1	0.00e+00	564
10	NC_069133.1	NC_047026.1	0.00e+00	527
11	NC_069138.1	NC_047016.1	0.00e+00	447
12	NC_069136.1	NC_047028.1	0.00e+00	428
13	NC_069130.1	NC_047030.1	0.00e+00	410
14	NC_069122.1	NC_047029.1	0.00e+00	371
15	NC_069131.1	NC_047032.1	0.00e+00	324
16	NC_069126.1	NC_047016.1	8.72e-184	305
17	NC_069128.1	NC_047033.1	0.00e+00	241
18	NC_069126.1	NC_047027.1	8.27e-230	223
19	NC_069137.1	NC_047031.1	1.95e-197	136
20	NC_069128.1	NC_047031.1	6.02e-04	27
21	NC_024738.1	NW_022980724.1	1.17e-08	3

e

Mare vs Pmax FET, chrom. pieces

	Mare scaf	Pmax scaf	p (FET)	Count
0	NC_069123.1 : 0-end	NC_047015.1 : 0-end	0.00e+00	958
1	NC_069132.1 : 0-end	NC_047017.1 : 0-end	0.00e+00	930
2	NC_069134.1 : 0-end	NC_047018.1 : 0-end	0.00e+00	824
3	NC_069127.1 : 0-end	NC_047019.1 : 0-end	0.00e+00	735
4	NC_069135.1 : 0-end	NC_047021.1 : 0-end	0.00e+00	685
5	NC_069129.1 : 0-end	NC_047025.1 : 0-end	0.00e+00	681
6	NC_069125.1 : 0-end	NC_047022.1 : 0-end	0.00e+00	623
7	NC_069124.1 : 0-end	NC_047020.1 : 0-end	0.00e+00	614
8	NC_069125.1 : 0-end	NC_047024.1 : 0-end	0.00e+00	592
9	NC_069124.1 : 0-end	NC_047023.1 : 0-end	0.00e+00	564
10	NC_069133.1 : 0-end	NC_047026.1 : 0-end	0.00e+00	527
11	NC_069138.1 : 0-end	NC_047016.1 : 0-end	0.00e+00	447
12	NC_069136.1 : 0-end	NC_047028.1 : 0-end	0.00e+00	428
13	NC_069130.1 : 0-end	NC_047030.1 : 0-end	0.00e+00	410
14	NC_069122.1 : 0-end	NC_047029.1 : 0-end	0.00e+00	371
15	NC_069131.1 : 0-end	NC_047032.1 : 0-end	0.00e+00	324
16	NC_069126.1 : 0-end	NC_047016.1 : 0-end	8.72e-184	305
17	NC_069128.1 : 0-end	NC_047033.1 : 0-end	0.00e+00	241
18	NC_069126.1 : 0-end	NC_047027.1 : 0-end	8.27e-230	223
19	NC_069137.1 : 0-end	NC_047031.1 : 0-end	1.95e-197	136
20	NC_069128.1 : 0-end	NC_047031.1 : 0-end	6.02e-04	27
21	NC_024738.1 : 0-end	NW_022980724.1 : 0-end	1.17e-08	3

(a) depicts the chromosomes/scaffolds of Mare (x-axis) plotted against the chromosomes/scaffolds of Pmax (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 12503 orthologs found between 18 Mare scaffolds and 212 Pmax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.